ELE101 Communication Systems

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Outline

- Review
  - Assignment submission
- Communication Systems
- Demo with SDR (Software Defined Radio)
- Python 101 cont'd

GitHub Classroom

1. Click on the assignment link
2. Authorize GitHub Classroom
3. Find yourself on the class roll
4. Match correctly, if you make error contact me!
5. Click on the "Accept this assignment" button.
6. Having accepted the assignment, a new repo will be created for you!.
7. Check your e-mail associated with GitHub for any pending invitation! Accept if any!
8. Read the README file to what to do!

Assignment Submission

- Clone assignment repo into your PC
- change to your local directory (using cd command)
- Follow instructions in the README file
- Submit your assignment using git commands
  - *git add .*
  - *git commit - m “something here”*
  - *git push*

How to use conda environment

- First, launch miniforge prompt
- Then, type
  - *conda env list*
  - *conda activate ELE101*
  - *jupyter-lab* ***or*** *jupyter lab*

Haberleşme / Bir çok soru(lar) ?

- Frekans / Spektrum Nedir ?
- HF, VHF, UHF ne anlama geliyor ?
- Modülasyon nedir ? FM, AM, …
- Şu “Karmaşık sayıları” nerede kullanıyorum ?
- Ses bozulmadan nasıl iletiliyor ?
- Hücresel / WiFi / GPS / IoT nasıl çalışır ?

Haberleşme

Bilginin karşı tarafa iletilmesi

Sinyal

- Akustik, borsa hisse, deprem, elektronik,
- Işık, radyo dalgaları: haberleşme/radar

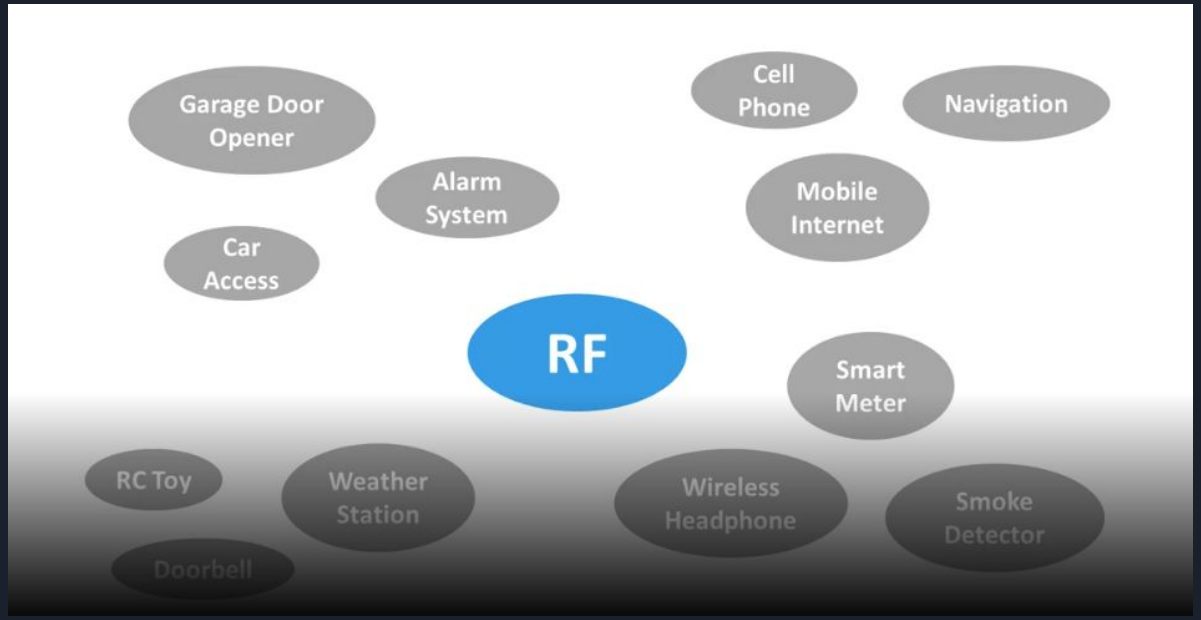
Farklı örnekler

- Mobil el telsizi
- FM Radyo istasyonu
- Nem ölçer

Kanal

- Hava
- Kablo

RF Her Yerde


Garage Door Opener
Cell Phone
Navigation
Alarm System
Mobile Internet
Car Access
RF
Smart Meter
RC Toy
Weather Station
Wireless Headphone
Smoke Detector
Doorbell

UNITED STATES FREQUENCY ALLOCATIONS

THE RADIO SPECTRUM

RADIO SERVICES COLOR LEGEND

- AERONAUTICAL MOBILE
- AERONAUTICAL MOBILE-SATELLITE
- AERONAUTICAL RADIONAVIGATION
- AMATEUR
- AMATEUR-SATELLITE
- BROADCASTING
- BROADCASTING SATELLITE
- EARTH EXPLORATION SATELLITE
- FIXED
- FIXED-SATELLITE
- INTER-SATELLITE
- LAND MOBILE
- LAND MOBILE SATELLITE
- MARITIME MOBILE
- MARITIME MOBILE SATELLITE
- MARITIME RADIONAVIGATION
- METEOROLOGICAL
- METEOROLOGICAL SATELLITE
- MOBILE
- MOBILE SATELLITE
- RADIO ASTRONOMY
- RADIODETERMINATION SATELLITE
- RADIOLOCATION
- RADIOLOCATION SATELLITE
- RADIONAVIGATION
- RADIONAVIGATION SATELLITE
- SPACE OPERATION
- SPACE RESEARCH
- STANDARD FREQUENCY AND TIME SIGNAL
- STANDARD FREQUENCY AND TIME SIGNAL SATELLITE

ACTIVITY CODE

- FEDERAL EXCLUSIVE
- FEDERAL NON-FEDERAL SHARED
- NON-FEDERAL EXCLUSIVE

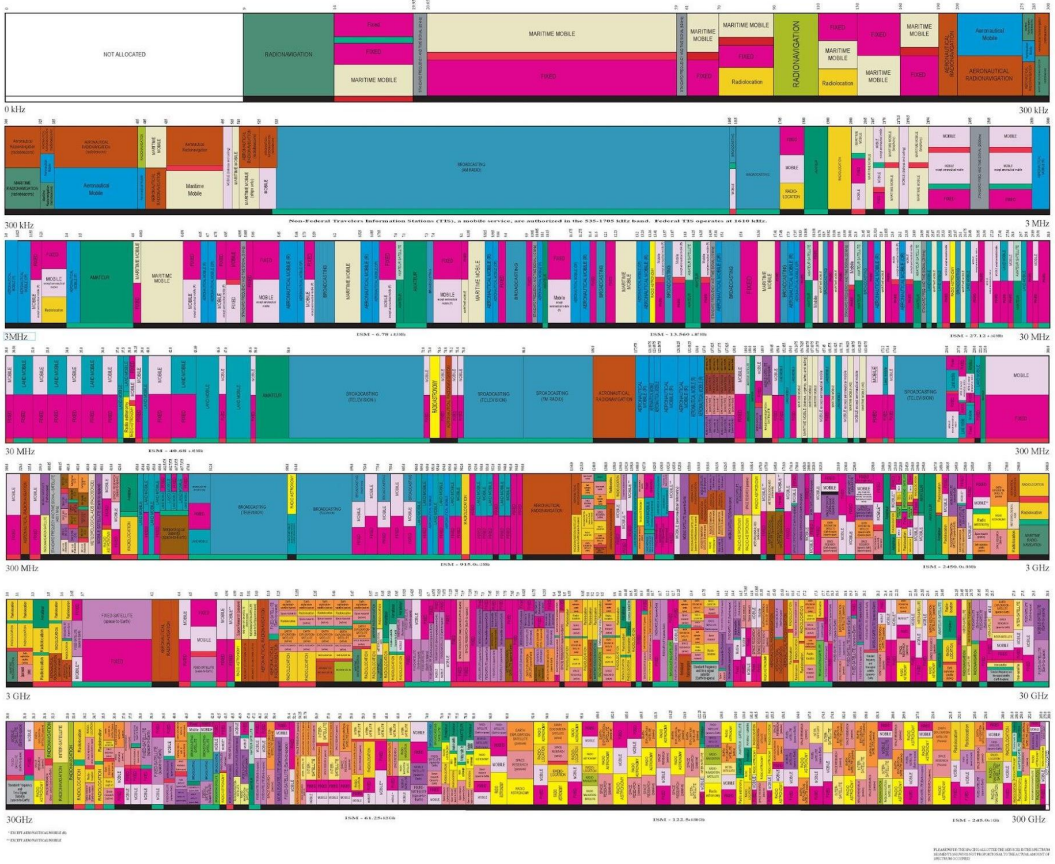
ALLOCATION USAGE DESIGNATION

SERVICE	EXAMPLE	DESCRIPTION
Primary	FIXED	Capital Letters
Secondary	Mobile	1st Capital with lower case letters

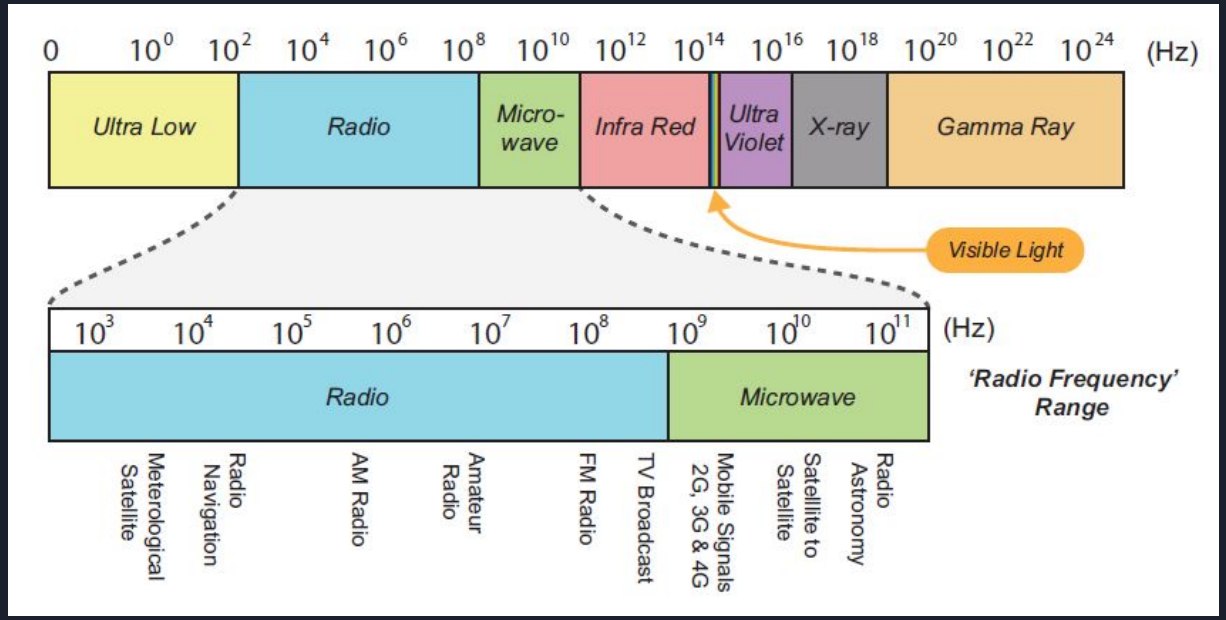
The charts on pages only represent the Table of Frequency Allocations used by the FCC and NTIA. As such, it may not emphasize other allocations, or footnotes, or recent changes made to the Table of Frequency Allocations. Therefore, for complete information, some details could be lost in attempts to convert some of U.S. allocations.

U.S. DEPARTMENT OF COMMERCE
National Telecommunications and Information Administration
Office of Spectrum Management
JANUARY 2016

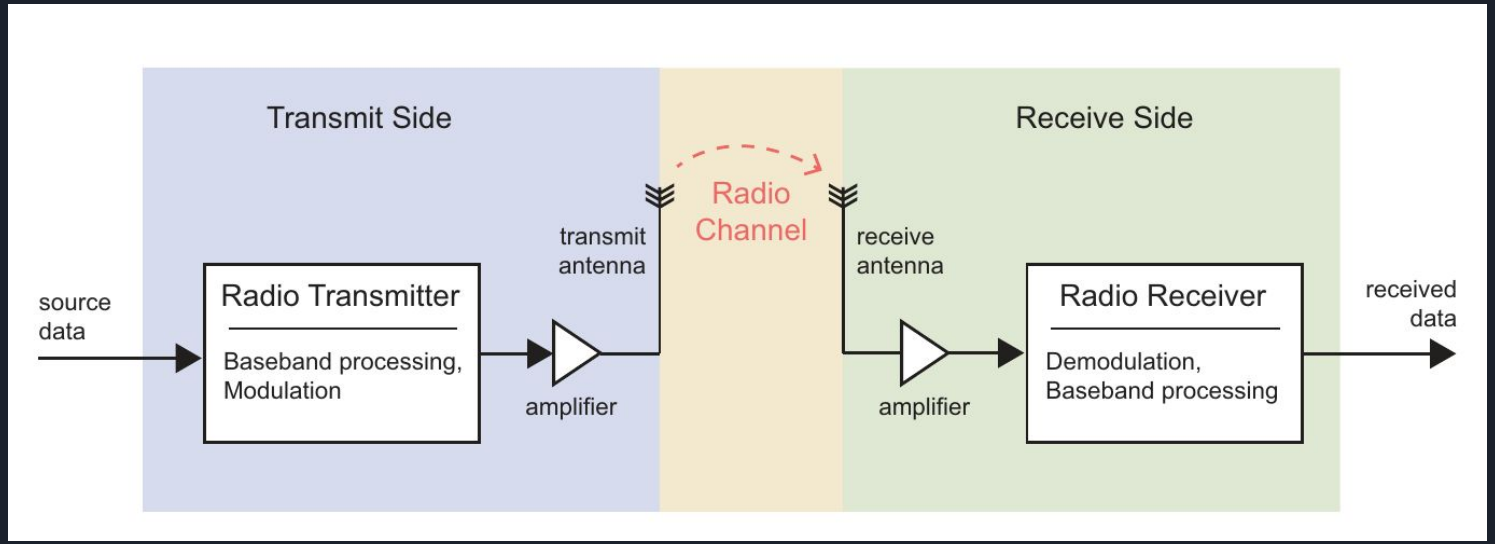
For sale by the Superintendent of Documents, U.S. Government Printing Office
Internet: bookstore.gpo.gov Phone: toll free (866) 512-1800; DC area (202) 512-1800
Fax: (202) 512-2250 Mail: Stop SSOP, Washington DC 20402-0001



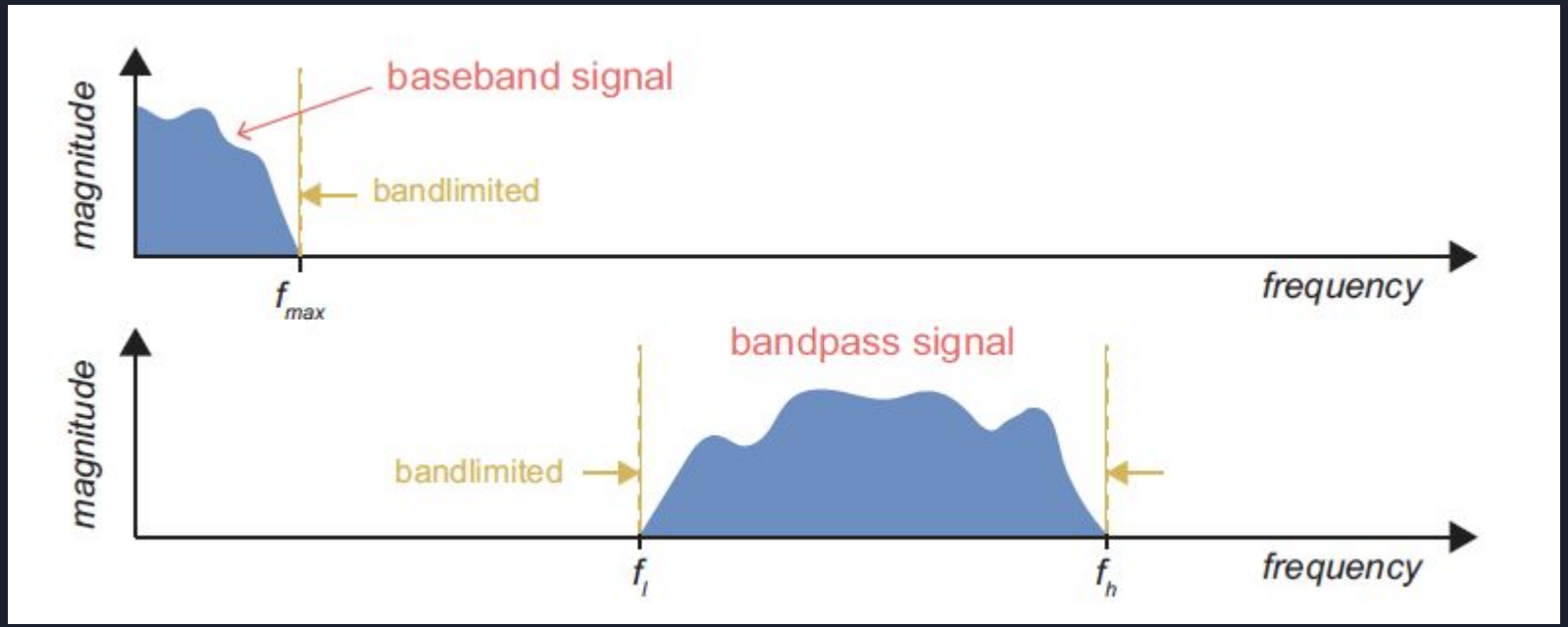
Elektromanyetik Spektrum



Basit Haberleşme Modeli

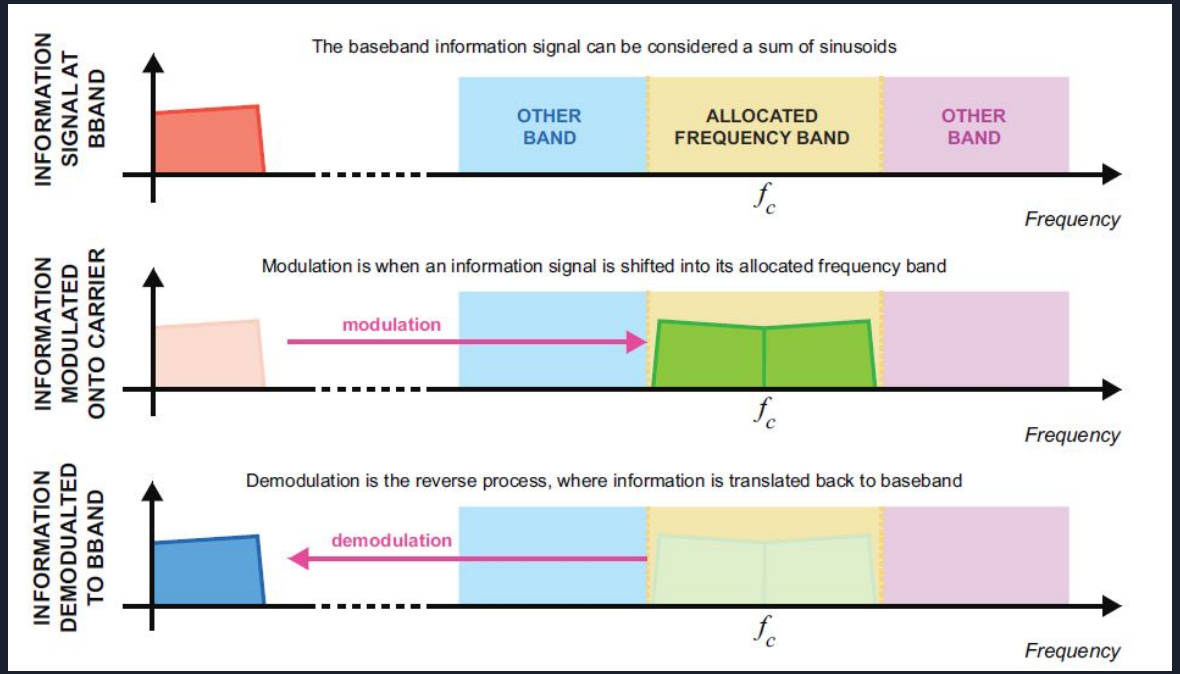


Frekans & Bant Genişliği



Modülasyon/Demodülasyon

MoDem = Modülatör + Demodülatör



Communication Systems

- Described as Band, Frequency, or Wavelength
- Bands: VHF
- Frequency: 100 MHz
- Wavelength: 3 m
- lambda * frequency = speed of light
  - $\lambda * f = c$
- Wavelength (in m) = 300 / (frequency in MHz)

5G mmWave Calculation

- 5G uses millimeter waves (mmWaves)
- mmWaves are broadcast at frequencies between 30 and 300 GHz
- So, what are the corresponding wavelengths?

Modülasyon Örnekleri

Analog

- AM
- FM
- PM

Sayısal

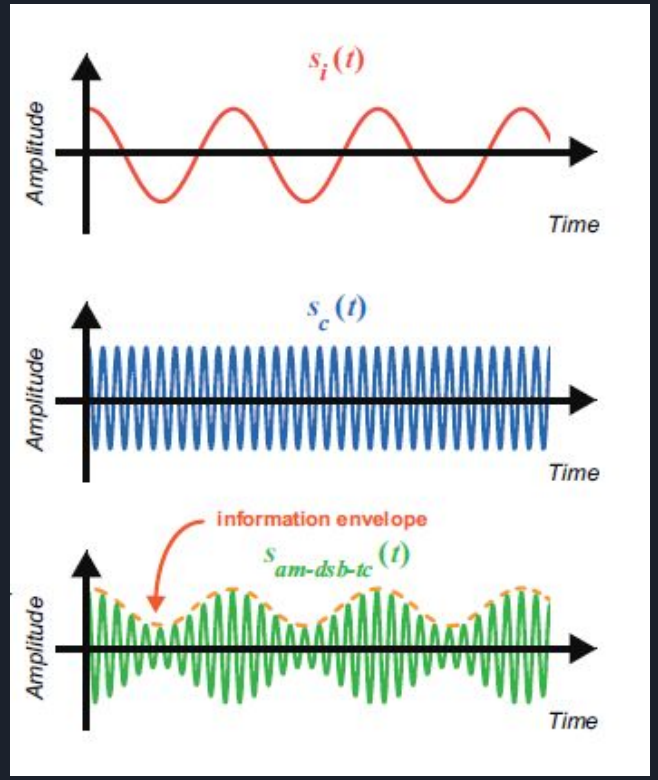
- OOK
- FSK
- QPSK/QAM16 ...

Amplitude Modulation (AM)
Genlik Kiplemesi

Taşıyıcı frekansın genliği değiştiriliyor.

Kompleks örneklenmiş verinin genliği bize bilgiyi veriyor.

Hava bandı konuşmaları

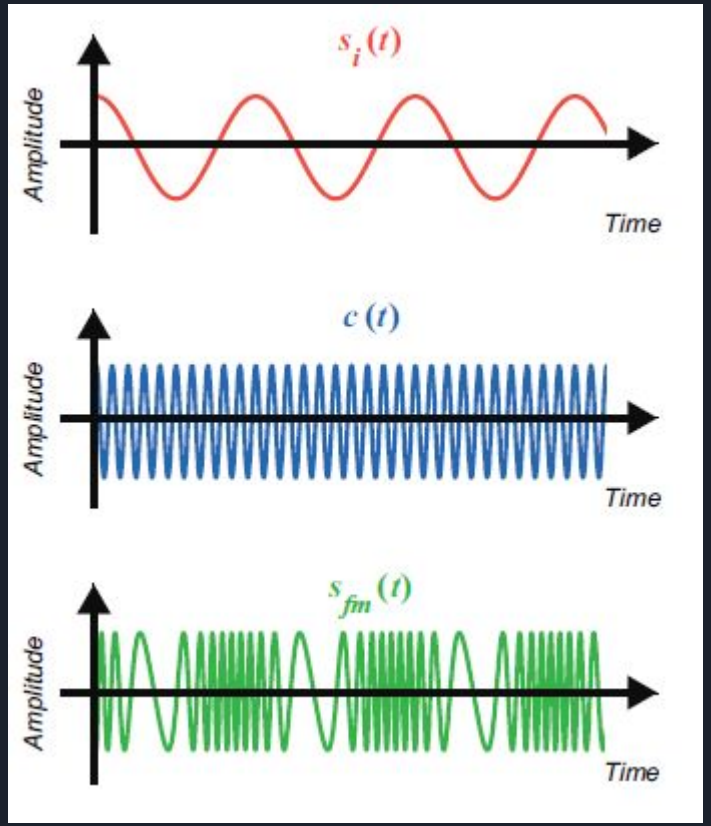


Frequency Modulation (FM)
Frekans Kiplemesi

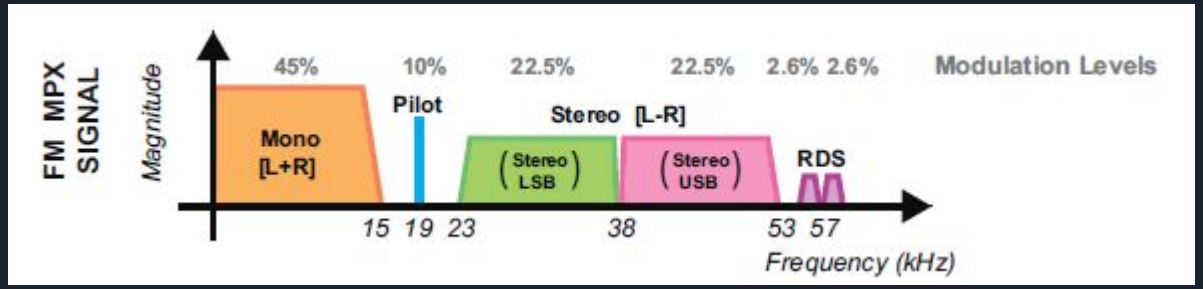
Taşıyıcı frekansın frekansı değiştiriliyor.

Kompleks örneklenmiş verinin türevi bize bilgiyi veriyor.

Gürültüye karşı daha dayanıklı



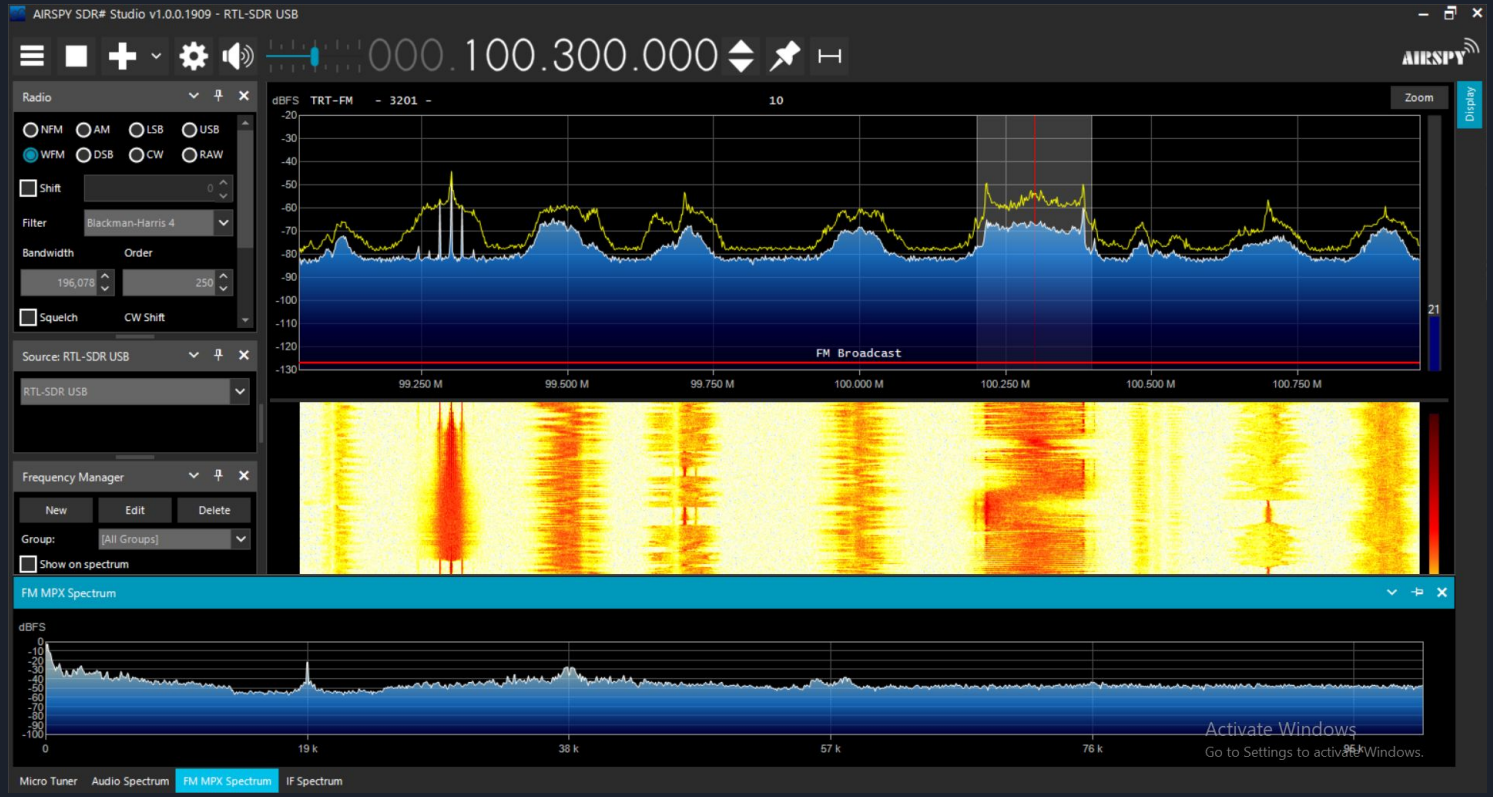
Geniş Bant Frekans Modülasyonu (WBFM)



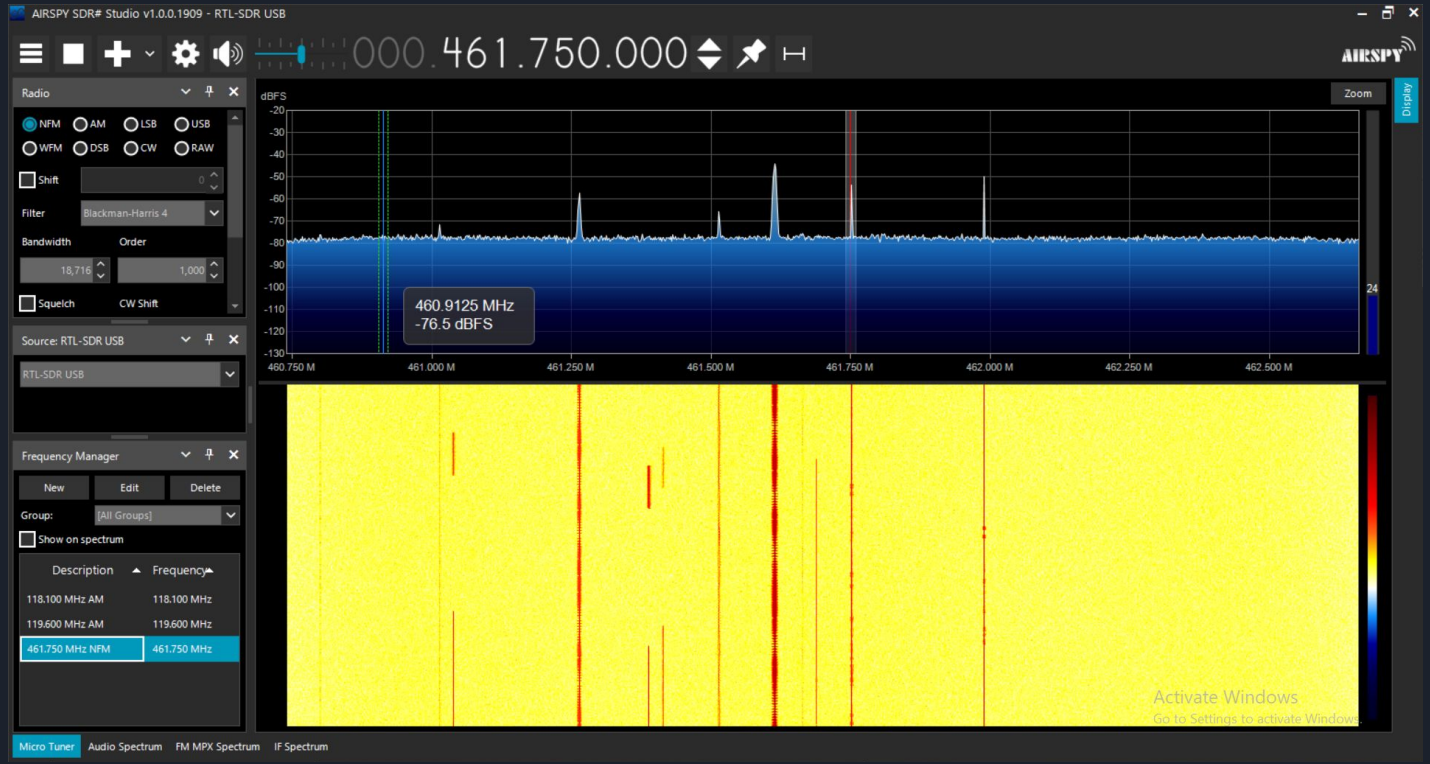
Yazılım Tabanlı Radyo

- Yazılım Tabanlı Radyo (YTR)
- Software Defined Radio (SDR) veya Software Radio (SR)
  - Systems Defined Radio
    - Elektromanyetik, RF, Haberleşme, DSP, Embedded, SW
- Nedir? (IEEE, Wireless Innovation Forum)
  - Bazı veya tüm fiziksel seviye fonksiyonları yazılım ile kontrol edilebilen radyo

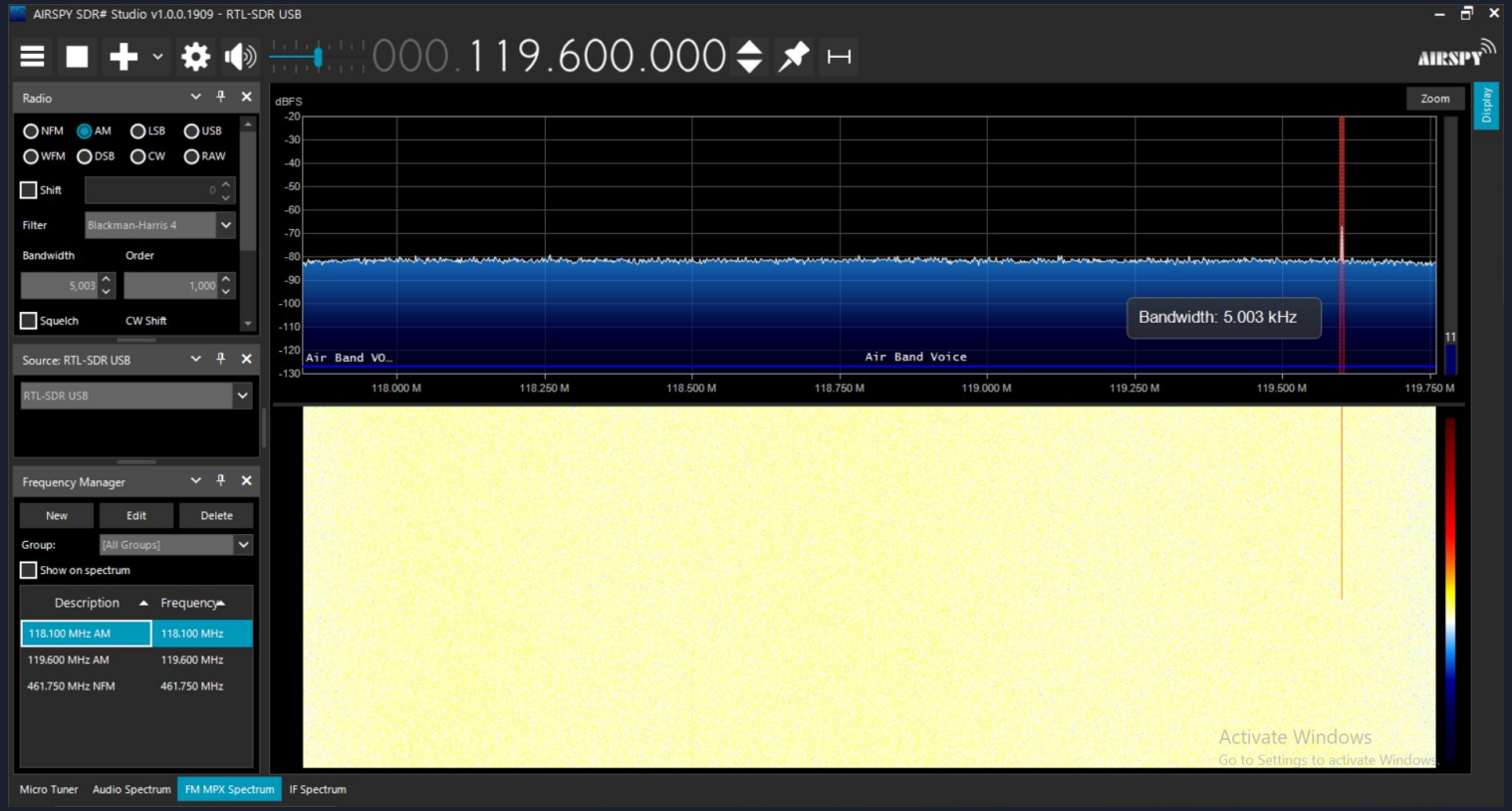
Merhaba Dünya: WBFM



NBFM


AIRSPY SDR# Studio v1.0.0.1909 - RTL-SDR USB
000.461.750.000
AIRSPY
Radio
NFM
AM
LSB
USB
WFM
DSB
CW
RAW
Shift
Filter
Blackman-Harris 4
Bandwidth
Order
18,716
1,000
Squelch
CW Shift
Source: RTL-SDR USB
RTL-SDR USB
Frequency Manager
New
Edit
Delete
Group:
[All Groups]
Show on spectrum
Description
Frequency
118.100 MHz AM
118.100 MHz
119.600 MHz AM
119.600 MHz
461.750 MHz NFM
461.750 MHz
dBFS
Zoom
Display
460.9125 MHz
-76.5 dBFS
460.750 M
461.000 M
461.250 M
461.500 M
461.750 M
462.000 M
462.250 M
462.500 M
Micro Tuner
Audio Spectrum
FM MPX Spectrum
IF Spectrum
Activate Windows

AM


AIRSPY SDR# Studio v1.0.0.1909 - RTL-SDR USB
000.119.600.000
AIRSPY
Radio
NFM
AM
LSB
USB
WFM
DSB
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Blackman-Harris 4
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1,000
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CW Shift
Source: RTL-SDR USB
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Show on spectrum
Description
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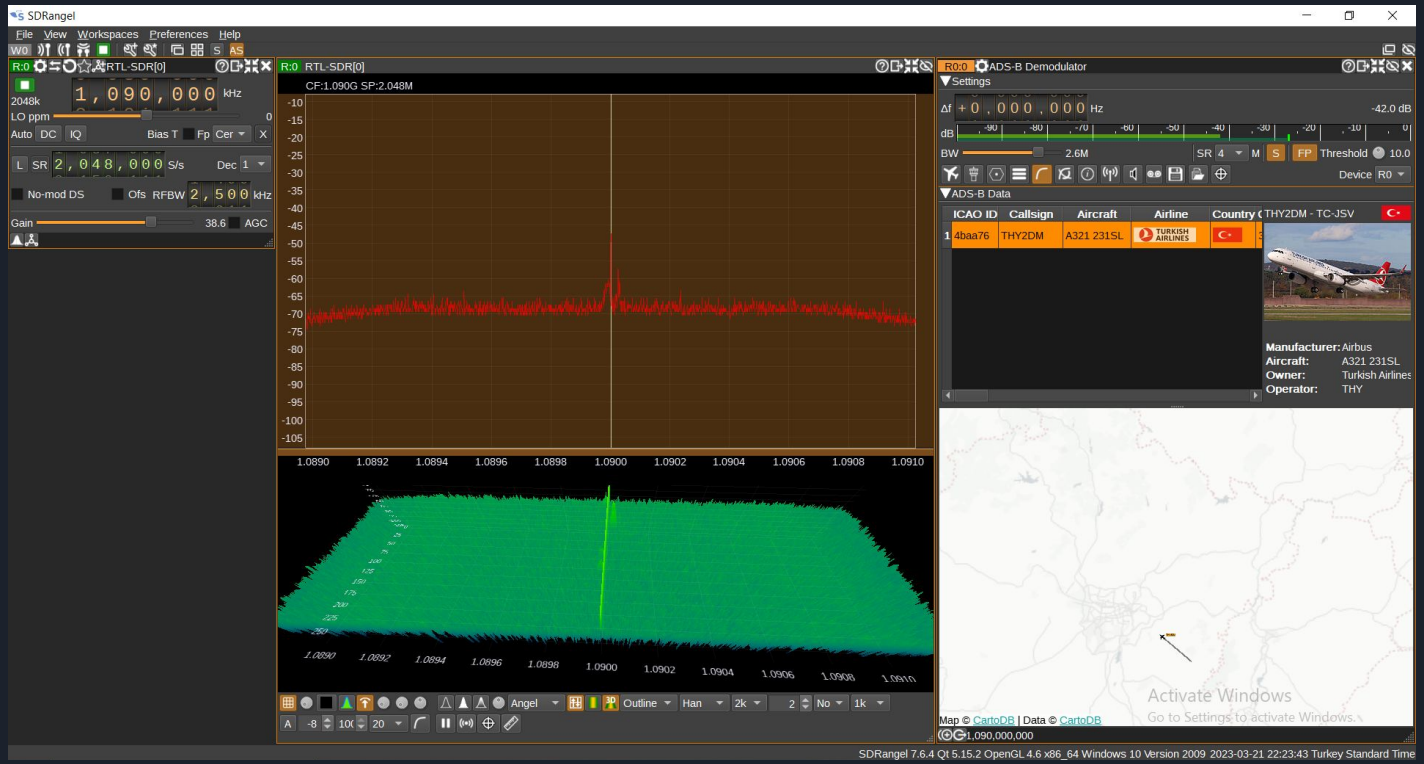
118.100 MHz AM
118.100 MHz
119.600 MHz AM
119.600 MHz
461.750 MHz NFM
461.750 MHz
dBFS
Zoom
Display
Bandwidth: 5.003 kHz
Air Band VO...
Air Band Voice
118.000 M
118.250 M
118.500 M
118.750 M
119.000 M
119.250 M
119.500 M
119.750 M
Micro Tuner
Audio Spectrum
FM MPX Spectrum
IF Spectrum
Activate Windows
Go to Settings to activate Windows.

ADS-B


SDRangel
File View Workspaces Preferences Help
RTL-SDR[0]
1,090,000 kHz
2048k
LO ppm
Auto DC IQ Bias T Fp Cer
SR 2,048,000 S/s Dec 1
No-mod DS Ofs RFBW 2,500 kHz
Gain 38.6 AGC
CF:1.090G SP:2.048M
ADS-B Demodulator
Settings
Δf +0,000,000 Hz
-42.0 dB
BW 2.6M
SR 4 M S FP Threshold 10.0
Device R0
ADS-B Data
ICAO ID Callsign Aircraft Airline Country
1 4baa76 THY2DM A321 231SL TURKISH AIRLINES
THY2DM - TC-JSV
Manufacturer: Airbus
Aircraft: A321 231SL
Owner: Turkish Airlines
Operator: THY
Map © CartoDB | Data © CartoDB
1,090,000,000
Activate Windows
Go to Settings to activate Windows.
SDRangel 7.6.4 Qt 5.15.2 OpenGL 4.6 x86_64 Windows 10 Version 2009 2023-03-21 22:23:43 Turkey Standard Time

Python101

- Open the notebooks

Thanks!

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